

Heterogeneous Agents in Macroeconomics

Paris-Dauphine & PSL Universities – M2 Quantitative Economics

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CHAPTER 1 Introduction to HA and OLG

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HANDOUT OBJECTIVES

- Build and calibrate an 80-cohort OLG economy where saving, capital, production, wages, and interest rates are jointly determined, using Dynare's macro-processor to generate it from a compact source file.
- Run and interpret perfect-foresight simulations of the calibrated economy.

General Introduction

The Lucas critique redefined what macroeconomics should be: models built on explicit preferences, technology, and expectations, evaluated as economic mechanisms rather than fitted curves (Lucas, 1976). Since then, macroeconomics has moved toward more stylized and simplified models. That simplicity was the price of internal consistency: tractable equilibrium logic, sharp identification of mechanisms, and clean policy experiments. On the other hand, it was a sacrifice. Aggregation requirements and the need for closed-form or near-analytical solutions pushed household heterogeneity out of the core framework and of the research agenda.

Since then, macro often starts from a representative household because aggregation is convenient: one Euler equation describes aggregate consumption, and one budget constraint summarizes household wealth. This representation is powerful when the conditions for aggregation hold. It becomes misleading when households differ precisely along the margins that determine their response to prices, income, risk, and policy. In some cases, the question itself simply cannot be addressed in a representative-agent economy.

The point of heterogeneity in macroeconomics is therefore not merely to acknowledge that agents are different. The question is whether the *distribution* of individual states is needed to determine aggregate quantities and equilibrium prices. Heterogeneity takes many forms. Firms and workers are spread across sectors linked by input-output networks. Workers differ by skills, tasks, and exposure to unemployment risk. Some households and firms borrow while others save. Agents differ by age and by the liquidity of their assets. If two economies have the same aggregate output, income, and wealth but a different sectoral composition, a different share of constrained borrowers, retirees, unemployed workers, or illiquid asset holders, do they react in the same way to a productivity shock, an interest-rate change, or a transfer? When the answer is no, the distribution is a state variable of the macroeconomic model. Incomplete-market economies with household wealth heterogeneity are the best-known branch, from the pure-exchange economy of Huggett (1993) and the production economy of Aiyagari (1994) to the New Keynesian extension of Kaplan et al. (2018). This course covers the broader family and asks, for each source of heterogeneity, which margin matters and what it changes at the aggregate level.

A cartography of heterogeneity. Because heterogeneity is a wide concept, Table 1 organizes the literature by the individual state that must be tracked. A sectoral model studies where production takes place (Horvath, 2000; Acemoglu et al., 2012). A labor model tracks skills, tasks, employment, or earnings risk (Acemoglu and Autor, 2011; Liu, 2026). A credit model distinguishes borrowers from lenders and liquid from illiquid wealth (Iacoviello, 2005; Kaplan et al., 2014). A life-cycle model uses age (Samuelson, 1958; Diamond, 1965; Auclert et al., 2021). An incomplete-market model follows the joint distribution of income and assets (Huggett, 1993; Aiyagari, 1994; Krusell and Smith, 1998). These dimensions can be combined, but each additional dimension enlarges the state space and changes the equilibrium mechanism.

Discrete versus continuous heterogeneity. A model with two household types is not intrinsically less economic than a model with a continuous wealth distribution. It answers a different question at a different resolution. TANK models, for example (Eggertsson and Krugman, 2012), isolate the contrast between optimizing savers and hand-to-mouth consumers. Whether aggregation survives heterogeneity is itself a quantitative question (Baqaee and Farhi, 2020). An annual OLG model uses a finite vector of cohorts to describe age. Aiyagari and HANK models typically combine discrete employment states with continuous asset holdings. Computation will eventually discretize a continuous state, but the underlying object remains a probability distribution.

The choice should follow the mechanism. If age is the main source of different horizons and saving motives, an OLG structure is natural. If liquidity is central, net worth is not enough and at least two

Dimension	Individual state	Discrete representation	Continuous representation	Examples
Sector / firm	sector, supplier, productivity, markup	a finite number of sectors or firm types	productivity and network exposure	Acemoglu et al. (2012) , Baqae and Farhi (2020)
Skills / tasks	education, occupation, task productivity	skilled/unskilled workers or occupations	multidimensional skills and task assignments	Acemoglu and Autor (2011) , Liu (2026)
Employment / income risk	employment, earnings, hours	employed / unemployed states	persistent earnings or productivity process	Aiyagari (1994) , Krusell and Smith (1998)
Borrower type / liquidity	debt, collateral, liquid and illiquid assets	borrower/saver, constrained/unconstrained, TANK	leverage, liquid wealth, loan-to-value ratios	Iacoviello (2005) , Eggertsson and Krugman (2012) , Kaplan et al. (2014)
Age / cohort	age, survival, efficiency, retirement	two-period OLG or annual cohorts	continuous age and survival hazards	Samuelson (1958) , Diamond (1965) , Auclert et al. (2021)
Wealth / portfolio	assets, portfolio composition, return type	wealth bins or hand-to-mouth classes	one- or two-asset distributions	Huggett (1993) , Kaplan et al. (2018) , Daminato and Pistaferri (2024)

Table 1: A practical map of heterogeneity in macroeconomics. Discrete and continuous representations are alternative approximations to an economic distribution, not mutually exclusive empirical claims.

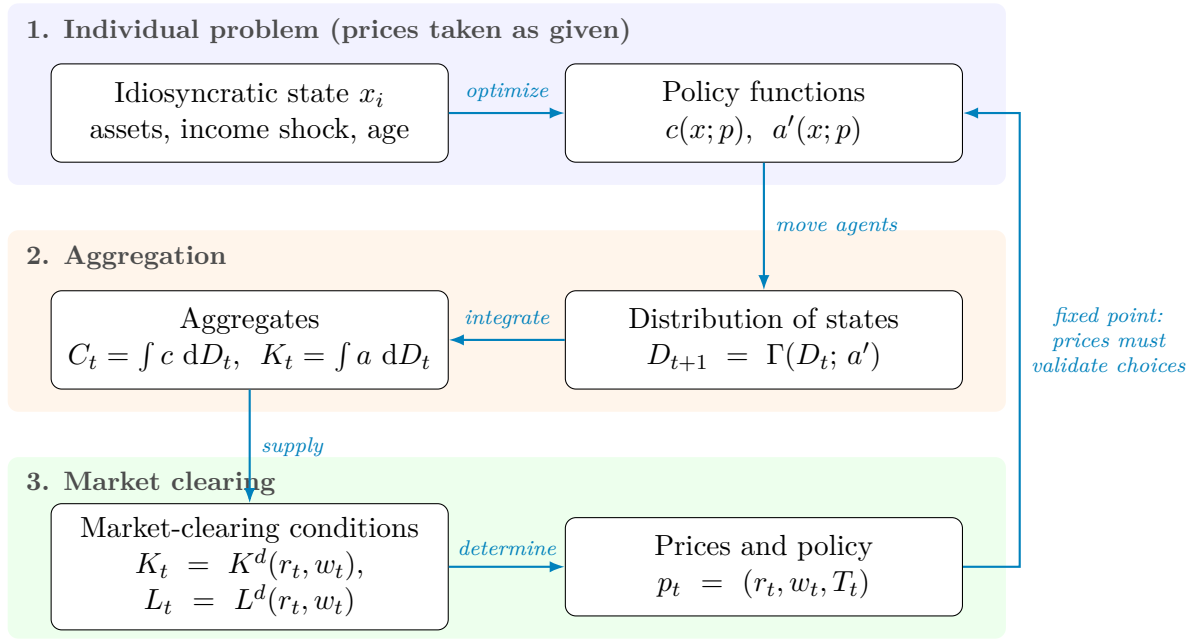
assets may be needed. If sectoral linkages transmit shocks, household age is secondary but the input-output network is essential. The model should retain the smallest distribution that changes the answer to the macroeconomic question.

From individual behavior to a macroeconomic equilibrium. Every heterogeneous-agent model contains the same feedback loop. Given prices, agents solve individual problems. Their policy rules move the distribution of states. Aggregation of that distribution produces consumption, labor, saving, and capital. Those aggregates determine prices, which must in turn validate the policy rules from which the aggregates were computed.

This chapter develops that loop progressively. The two-agent economy makes market clearing visible with almost no computational machinery. The N -agent economy shows how sums become integrals over a distribution. The OLG economy then provides a quantitative application in which 79 cohort-level Euler equations, 80 budget constraints, production, and capital-market clearing must be solved together.

The three-session course: The course asks when household heterogeneity changes a macroeconomic answer, and how to compute the equilibrium that answers that question. We do not add every friction to one ever-growing model. We retain the household–distribution–aggregation–prices loop and deliberately change the source of heterogeneity and the asset-market closure:

1. **Session 1: market clearing and life-cycle heterogeneity.** Following [Samuelson \(1958\)](#) and [Diamond \(1965\)](#), two-agent examples lead to an 80-cohort OLG production economy. Predictable age-dependent earnings and remaining horizons generate different saving profiles. We use Dynare’s macro-processor, nonlinear perfect-foresight paths, and balanced redistribution with and without borrowing limits.
2. **Session 2: uninsured income risk and HA-RBC.** In the tradition of [Huggett \(1993\)](#) and [Aiyagari \(1994\)](#), infinitely lived households face idiosyncratic earnings shocks and incomplete insurance. Aiyagari already has productive capital and endogenous factor prices. We compute policies, the stationary distribution and equilibrium or calibration, then study a small aggregate productivity shock. The Krusell–Smith-style extension ([Krusell and Smith, 1998](#)) adds aggregate risk, not capital.



Note: (1) Given prices, each agent chooses from its idiosyncratic state. (2) These choices move the distribution, whose integrals give the aggregates. (3) Markets clear and determine prices. In equilibrium, the prices used in step 1 are those produced in step 3.

Figure 1: The heterogeneous-agent fixed point in three layers.

3. Session 3: HANK and policy experiments. We retain income risk and incomplete insurance but replace capital with government bonds and introduce sticky nominal wages, following Pieroni’s sticky-wage HANK and in the spirit of [Kaplan et al. \(2018\)](#). We compare monetary transmission with RANK, then change fiscal financing and transfer targeting in a working Dynare laboratory.

Age, uninsured earnings risk and nominal rigidity are thus examined separately before students attempt richer combinations.

Common conventions and software. Throughout the course a_t and K_t denote end-of-period choices, so inherited assets and installed capital carry a lag. We use D_t for the distribution, e for individual labor efficiency, Z for aggregate productivity, r for a net real return and σ for inverse EIS. A shared notation sheet at the end of every handout records code aliases and model-specific objects. Periods are annual in Session 1, illustrative model periods in Session 2 and quarterly in Session 3: common symbols do not imply identical calibration values or units.

Install **Dynare 7.x**, at least **7.1**; **version 7.2 is recommended**, with MATLAB. The HA blocks in Sessions 2–3 must not be run with 7.0 because of its complementarity-sign bug. Python is used for their figure and validation scripts, not as a replacement equilibrium solver.

PART I : Heterogeneity Challenges

With heterogeneous households, individual optimality no longer describes aggregate equilibrium by itself. Each Euler equation gives a desired saving position at a candidate return. The equilibrium return is

the price that makes all these desired positions compatible with the assets actually available. The Euler equation and the market-clearing condition therefore play different roles, and heterogeneity is what makes the difference visible.

Section A shows this in a two-period endowment economy, first with one household and then with two households that differ in patience. Section B extends the argument to N households, to productive capital, and to a distribution of individual states. This last step is a bridge toward HANK models, but it is not yet a HANK model.

A. A Two-Agent Economy

The representative-agent benchmark. Consider first a two-period endowment economy. There is no uncertainty in this example: e_1 and e_2 are deterministic endowments known from the outset, not two successive shocks. One household receives (e_1, e_2) , values consumption according to

$$u(c_1, c_2) = \log c_1 + \beta \log c_2,$$

and can buy a one-period asset a that pays the net return r :

$$c_1 + a = e_1, \tag{1}$$

$$c_2 = e_2 + (1 + r)a. \tag{2}$$

Substituting the constraints into utility reduces the problem to choosing saving:

$$\max_a \log(e_1 - a) + \beta \log(e_2 + (1 + r)a).$$

The first-order condition is the familiar Euler equation,

$$\frac{1}{c_1} = \beta(1 + r) \frac{1}{c_2}. \tag{3}$$

Equation (3) describes an optimal response to a *given* return. It is not yet an equilibrium. To determine r , one must specify who issues the asset and how much of it is available.

Suppose the asset is a private bond in zero net supply. With a single household, market clearing imposes $a = 0$. Hence $c_1 = e_1$, $c_2 = e_2$, and the equilibrium return is

$$1 + r^* = \frac{e_2}{\beta e_1}. \tag{4}$$

The rate does not appear because a planner mechanically assigns consumption. It appears because a price must make the household voluntarily choose the only feasible aggregate asset position. Even in the representative-agent benchmark, the Euler equation and the market-clearing condition play different roles.

Budget	$c_1 + a = e_1, c_2 = e_2 + (1 + r)a$	Euler equation	$\frac{1}{c_1} = \beta(1 + r) \frac{1}{c_2}$
Bond clearing	$a = 0$	Equilibrium	$c_1 = e_1, c_2 = e_2, 1 + r^* = \frac{e_2}{\beta e_1}$

Two households. Now let $i \in \{A, B\}$ index two households with possibly different endowments and discount factors. Household i solves

$$\max_{a_i} \log(e_{1i} - a_i) + \beta_i \log(e_{2i} + (1 + r)a_i).$$

Its Euler equation is

$$\frac{1}{e_{1i} - a_i} = \beta_i(1 + r) \frac{1}{e_{2i} + (1 + r)a_i}.$$

Solving this equation for desired asset holdings gives

$$a_i(r) = \frac{\beta_i(1 + r)e_{1i} - e_{2i}}{(1 + r)(1 + \beta_i)}. \quad (5)$$

This expression is an *individual demand schedule*. It tells us whether household i wishes to lend or borrow at each possible return. It does not tell us which return is feasible for the economy.

With the bond in zero net supply, equilibrium requires

$$a_A(r^*) + a_B(r^*) = 0, \quad (6)$$

Substituting equation (5) into (6) yields

$$1 + r^* = \frac{\sum_{i \in \{A, B\}} \frac{e_{2i}}{1 + \beta_i}}{\sum_{i \in \{A, B\}} \frac{\beta_i e_{1i}}{1 + \beta_i}}. \quad (7)$$

Budget	$c_{1i} = e_{1i} - a_i, c_{2i} = e_{2i} + (1 + r)a_i$	Euler equation	$\frac{1}{c_{1i}} = \beta_i(1 + r) \frac{1}{c_{2i}}$
Asset demand	$a_i(r) = \frac{\beta_i(1 + r)e_{1i} - e_{2i}}{(1 + r)(1 + \beta_i)}$	Bond clearing	$a_A(r^*) + a_B(r^*) = 0$
Equilibrium return	$1 + r^* = \frac{\sum_{i \in \{A, B\}} \frac{e_{2i}}{1 + \beta_i}}{\sum_{i \in \{A, B\}} \frac{\beta_i e_{1i}}{1 + \beta_i}}$		

Equation (7) is a general-equilibrium result, not an implication of either household's Euler equation taken separately. At every candidate return, equations (5) give two individual saving demands. The equilibrium return r^* is the price that makes their joint asset holdings sum to zero. When discount factors differ, the equilibrium rate depends on the distribution of endowments and patience, not only on aggregate resources.

Example 1 (A patient lender and an impatient borrower) Take

$$(e_{1A}, e_{2A}, \beta_A) = (1, 1, 0.95), \quad (e_{1B}, e_{2B}, \beta_B) = (1, 1, 0.70).$$

Equation (7) gives $1 + r^* = 1.225$. Household A lends $a_A = 0.068$ and household B borrows $a_B = -0.068$. Both households receive exactly the same deterministic endowment in both periods. Trade arises solely because the more patient household wants to move consumption toward the future while the less patient household wants to bring consumption forward. The example therefore needs no anticipated income shock to generate a non-trivial asset market.

Household	e_1	e_2	β	$a(r^*)$	c_1	c_2
Patient A	1.00	1.00	0.95	0.068	0.932	1.084
Impatient B	1.00	1.00	0.70	-0.068	1.068	0.916
Aggregate	2.00	2.00		0	2.000	2.000

Table 2: Two-agent equilibrium. Individual Euler equations determine desired saving at a given return, and the market-clearing equation determines the return.

Why aggregation can sometimes work. If the two households have the same β and log utility, equation (7) simplifies to

$$1 + r^* = \frac{e_{2A} + e_{2B}}{\beta(e_{1A} + e_{1B})}.$$

Only aggregate endowments matter for the equilibrium rate. Yet individual bond positions can remain different when the timing of individual income differs. This is an important benchmark: heterogeneity exists, but it does not necessarily affect all aggregates. Representative-agent aggregation fails when the distribution changes aggregate excess demand at given prices, for example through heterogeneous patience, borrowing limits, uninsured risks, non-homothetic preferences, or different marginal propensities to consume. In Table 2, income timing is deliberately held fixed, so the entire credit position comes from heterogeneous patience.

B. Generalization

From two to N agents. Let $i = 1, \dots, N$ and allow each household to have $(e_{1i}, e_{2i}, \beta_i)$. The log-utility economy still has a closed-form equilibrium:

$$1 + r_N^* = \frac{\sum_{i=1}^N \frac{e_{2i}}{1 + \beta_i}}{\sum_{i=1}^N \frac{\beta_i e_{1i}}{1 + \beta_i}}. \quad (8)$$

With more general preferences, closed forms disappear, but the equilibrium definition is unchanged: the return r^* must make the sum of the households' desired asset holdings $a_i(r)$ equal to zero,

$$\sum_{i=1}^N a_i(r^*) = 0. \quad (9)$$

The two-agent example is the special case $N = 2$. Numerically, one evaluates all household policy rules at a candidate return, aggregates desired saving, and changes the return until net asset demand is zero.

The computational burden grows for two distinct reasons. First, there are more individual problems. Second, and more importantly, the distribution of individual states evolves over time. Current prices affect saving. Saving changes tomorrow's wealth distribution. Tomorrow's distribution affects aggregate saving and future prices. Solving a transition therefore requires an entire sequence of prices that is consistent with the induced sequence of distributions.

From a finite population to a distribution. In the two-period economy, household i is fully described by its endowments and its discount factor. Call this vector its *state*, $x_i = (e_{1i}, e_{2i}, \beta_i)$. Two households with the same state make the same choice, so desired saving can be written $a(x_i; r)$. Market clearing (9) can then be computed by grouping households with the same state rather than listing them one by one. Let $\mathcal{X}_N$ be the finite set of distinct states present in the population, and $D^N(x)$ the share of the N households whose state is $x \in \mathcal{X}_N$. Dividing (9) by N gives

$$\frac{1}{N} \sum_{i=1}^N a(x_i; r^*) = \sum_{x \in \mathcal{X}_N} D^N(x) a(x; r^*) = 0. \quad (10)$$

D^N is a discrete probability distribution, a histogram of the population across states: its weights are non-negative and sum to one.

When $N \rightarrow \infty$, the shares $D^N(x)$ converge to a distribution D on the set $\mathcal{X}$ of all possible states, and the weighted sum becomes an integral over $\mathcal{X}$,

$$\int_{\mathcal{X}} a(x; r^*) \, dD(x) = 0.$$

If D has a density f , this integral is $\int_{\mathcal{X}} a(x; r^*) f(x) \, dx$, the continuous version of the weighted sum. The notation $dD(x)$ is kept because the distributions of heterogeneous-agent models often have no density. Households at a borrowing limit, for instance, form a mass point. In practice, HANK models are solved on a finite grid of states, so the computed distribution is again a histogram, exactly as in (10). The continuum is a modeling device. With a continuum of households, idiosyncratic risk averages out, so the distribution and the aggregates are deterministic for a given path of prices.

A distribution that moves over time. In dynamic models the state $x_{i,t}$ changes every period. It contains the assets brought into the period and current idiosyncratic variables such as employment, age, or productivity. Given the prices p_t (the interest rate, the wage, taxes), each household chooses next-period assets according to a policy rule $a'(x; p_t)$. These choices, together with idiosyncratic shocks, move households across states and define the law of motion of the distribution,

$$D_{t+1} = \Gamma(D_t; a'), \quad (11)$$

which is the middle block of Figure 1. With productive capital, the asset is no longer in zero net supply. Household saving finances the capital stock, and the clearing condition becomes

$$\int_{\mathcal{X}} a'(x; p_t) \, dD_t(x) = K_t. \quad (12)$$

The population has mass one, so K_t is capital per household, the counterpart of $\frac{1}{N} \sum_{i=1}^N a'_{i,t}$ in the finite economy.

With production, aggregate saving must equal firms' demand for capital. In general, (r_t, w_t) must be determined together so household choices and both factor markets are consistent. Here labor supply is fixed at $\bar{L}$, so the wage follows from firms' labor-demand condition and r_t adjusts until household assets equal K_t .

The OLG economy below uses the same normalization: the cohort weights satisfy $\sum_j N_j = 1$, so cohort-weighted sums are per-capita quantities, not unweighted totals.

Equations (11) and (12) summarize the central difficulty: prices determine policies, policies determine the next distribution, and the distribution must generate the aggregates that justify prices.

The link with the first quantitative heterogeneous-agent models. The progression above is the stripped-down version of the research program that emerged in the late 1980s and 1990s. [İmrohoroglu \(1989\)](#) studied a general-equilibrium economy with idiosyncratic employment risk, incomplete insurance, indivisible employment, and a liquid asset in order to measure the welfare cost of business cycles. [Huggett \(1993\)](#) then placed the equilibrium risk-free rate at the center of a pure-exchange heterogeneous-agent economy. His stationary market-clearing condition is the continuum counterpart of equation (6):

$$\int_{\mathcal{X}} a(x; r^*) \, dD(x) = 0.$$

The interest rate must make the joint desired saving of the entire distribution equal the zero net supply of the risk-free asset. [Aiyagari \(1994\)](#) changed the closure by letting household saving finance productive capital. Asset-market clearing becomes $\int_{\mathcal{X}} a(x) \, dD(x) = K$, and the return is tied to the marginal product of capital. Finally, [Krusell and Smith \(1998\)](#) introduced aggregate productivity risk. The wealth distribution then becomes part of the aggregate state because households must forecast how it affects future prices. The sequence Huggett–Aiyagari–Krusell–Smith therefore mirrors the logic developed here: zero-supply bond clearing, productive-capital clearing, and then a distribution evolving under aggregate uncertainty. Ayşe İmrohoroglu’s contribution provides an earlier quantitative bridge from incomplete insurance and individual risk to aggregate welfare questions.

PART II : An OLG Economy

Age provides a structured way to move from an abstract distribution of agents to a quantitative heterogeneous-agent model. Cohorts differ in labor income, accumulated wealth, remaining lifetime, retirement status, and exposure to policy. These differences generate saving and consumption profiles even when all households share the same preferences. They also make the demographic composition of the population an equilibrium object: aging changes the mass of savers and dissavers and therefore affects capital and the interest rate.

This part separates the economic model from the evidence used to discipline it. Section A presents the model in the standard order (households, firms, and aggregation). Section B discusses the calibration, the computation of the stationary equilibrium and the data. Section C maps these blocks into Dynare and explains the macro-processor. Section D studies nonlinear perfect-foresight transitions. The objective is to make every code block traceable to an economic agent or a market-clearing condition.

A. Model

OLG models originate with [Samuelson \(1958\)](#), who studied intergenerational exchange with finite lives, and [Diamond \(1965\)](#), who introduced productive capital and public debt. Quantitative OLG models became central to fiscal and demographic analysis with [Auerbach and Kotlikoff \(1987\)](#). The household block follows the life-cycle hypothesis of [Modigliani and Brumberg \(1954\)](#) and [Ando and Modigliani \(1963\)](#): households smooth consumption over a finite life, so they save while working and run down their assets in retirement. The model below keeps this architecture deliberately small in order to isolate cohort accounting and capital-market clearing.

Demography. Time is annual and the economy is populated by $J = 80$ overlapping cohorts indexed by age $j = 1, \dots, J$, corresponding to ages 20–99. At each date a new cohort enters and the oldest one exits, so households of all ages coexist. Calendar time and age move together: a household of age j at date t is of age $j + 1$ at date $t + 1$. Let s_j denote the probability of surviving from age j to age $j + 1$, and let $d_j = 1 - s_j$ be the corresponding death probability. The probability of reaching age j from entry is

$$S_1 = 1, \quad S_j = \prod_{h=1}^{j-1} s_h. \quad (13)$$

Let $N_{j,t}$ be the mass of age- j households at date t . Aging shifts survivors into the next cohort:

$$N_{j+1,t+1} = s_j N_{j,t}, \quad j = 1, \dots, J - 1. \quad (14)$$

Births determine $N_{1,t+1}$. If population grows at rate n , normalized cohort shares satisfy $N_{j+1} = s_j N_j / (1 + n)$. We normalize total population to one, $\sum_{j=1}^J N_j = 1$, so that each N_j is simply the weight of cohort j and aggregates are weighted averages of cohort variables. Survival probabilities thus determine both individual horizons and these weights. The pedagogical code adopts the transparent limiting case $s_j = 1$ for $j < J$ and $s_J = 0$: households live with certainty through age 99 and then die. With zero population growth this produces equal cohort masses, $N_j = 1/J$. A richer demographic calibration would replace these values with age-specific survival rates.

Households. Households differ only by age. Labor efficiency e_j is an exogenous age profile. It is hump-shaped over the working life and equal to zero from the retirement age $J_R = 46$ onward, that is 65 years since age $j = 1$ is 20 years. Retirees therefore live on their accumulated assets. Let $a_{j,t}$ denote assets held at the end of date t by the age- j cohort. Taking prices (r_t, w_t) as given, a household born at date t solves

$$\max_{\{c_{j,t+j-1}\}_{j=1}^J, \{a_{j,t+j-1}\}_{j=1}^{J-1}} \sum_{j=1}^J \beta^{j-1} S_j \frac{c_{j,t+j-1}^{1-\sigma} - 1}{1 - \sigma} \quad (15)$$

subject to the budget constraints, written for age j at date t (a household born at t_0 is of age j at date $t_0 + j - 1$),

$$c_{j,t} + a_{j,t} = (1 + r_t) a_{j-1,t-1} + w_t e_j, \quad j = 1, \dots, J - 1, \quad (16)$$

$$c_{J,t} = (1 + r_t) a_{J-1,t-1} + w_t e_J, \quad (17)$$

with $a_{0,t-1} = 0$.¹

Optimal intertemporal consumption satisfies

$$c_{j,t}^{-\sigma} = \beta s_j (1 + r_{t+1}) c_{j+1,t+1}^{-\sigma}, \quad j = 1, \dots, J - 1. \quad (18)$$

The return is dated $t + 1$ because assets chosen at the end of t finance consumption when the household is one year older. The survival factor lowers the value of transferring resources to ages that may not be reached. The current code sets $s_j = 1$ for every interior Euler equation. Equation (18) then reduces exactly to the condition implemented in Dynare.

¹The objective is the expected lifetime utility of a household that is still alive at age j with probability S_j . Survival therefore acts like additional discounting: consumption at older ages matters only in states in which the household is alive. Newborn households enter without assets ($a_{0,t-1} = 0$), and the household chooses at each age how much to consume and how much to save in the one-period asset. Resources at age j are the return on inherited assets and labor income $w_t e_j$. The last cohort leaves no assets and consumes all remaining resources, as in equation (17).

Firms. Competitive firms operate a Cobb–Douglas technology,

$$Y_t = Z_t K_{t-1}^\alpha L_t^{1-\alpha}, \quad (19)$$

Capital installed in the previous period produces at date t . Aggregate productivity Z_t is total factor productivity, an exogenous AR(1) process in logs, $\log Z_t = \rho_Z \log Z_{t-1} + \varepsilon_t^Z$, with innovation $\varepsilon_t^Z \sim \mathcal{N}(0, \sigma_Z^2)$. The depreciation rate δ_t is also exogenous and mean-reverting, $\delta_t = (1 - \rho_D)\delta + \rho_D \delta_{t-1} + \varepsilon_t^D$: a positive innovation destroys part of the capital stock's return, as in a financial crisis. Flexible-price factor markets imply

$$w_t = (1 - \alpha) \frac{Y_t}{L_t}, \quad (20)$$

$$r_t = \alpha \frac{Y_t}{K_{t-1}} - \delta_t. \quad (21)$$

There is no New Keynesian block: no price dispersion, Phillips curve, nominal interest rate, or Taylor rule. The real interest rate is the marginal product of capital net of depreciation.

Aggregation and equilibrium. Aggregate consumption and end-of-period capital are

$$C_t = \sum_{j=1}^J N_j c_{j,t} \quad \text{and} \quad K_t = \sum_{j=1}^{J-1} N_j a_{j,t}. \quad (22)$$

Labor supply is fixed, so the labor market clears when firms' labor input equals aggregate efficiency units,

$$\bar{L} = L_t = \sum_{j=1}^J N_j e_j. \quad (23)$$

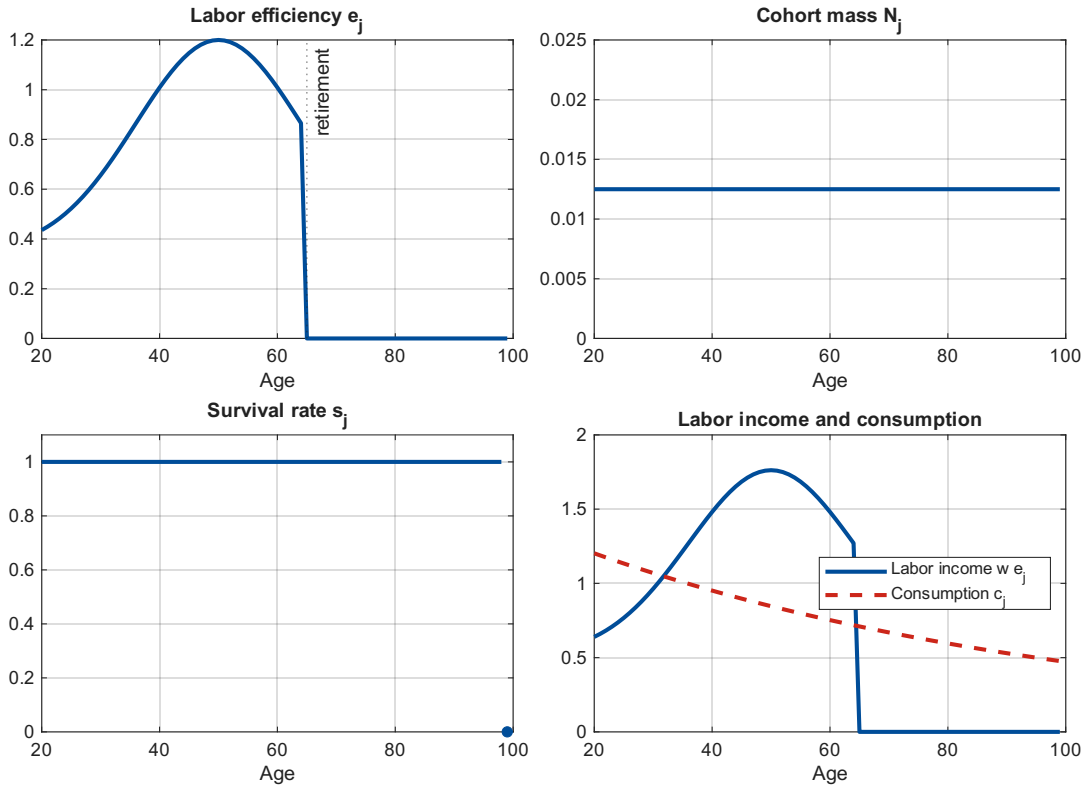
Equation (22) is the bridge between household saving and production. If cohorts wish to save more at given prices, capital cannot remain fixed: the extra saving changes K_t , future output, wages, and returns. The perfect-foresight solver searches for all cohort allocations and the entire price path that satisfy the household equations, firm equations, and market clearing simultaneously.

B. Calibration and the Stationary Equilibrium

Calibration. The model has two kinds of parameters. A handful of scalars are common to all households and firms: preferences (β, σ) , technology (α, δ) and the shock processes. Their values are standard annual choices, collected in Table 3. The other parameters are *heterogeneous*: one value per age for labor efficiency e_j , cohort masses N_j and survival rates s_j , that is 3×80 numbers. Listing them in a table would hide their economic content. Figure 2 displays them as age profiles instead. The efficiency profile is a hump peaking at age 50 and falls to zero at retirement, survival equals one until the terminal age, and the masses are equal because survival is certain and population does not grow. The fourth panel anticipates the stationary equilibrium computed below. Because consumption is smooth while income is hump-shaped, young households consume more than they earn and borrow. This is why the borrowing limit matters later.

Parameter	Value	Interpretation
β	0.97	annual discount factor
σ	2	inverse EIS / risk aversion
α	0.33	capital share
δ	0.06	annual depreciation
ρ_Z	0.9	TFP persistence
σ_Z	0.01	TFP innovation standard deviation
ρ_D	0.9	depreciation persistence
ε_1^D	0.02	initial depreciation innovation (financial crisis)
β^{RA}	$1/(1 + r^*) = 0.993$	RANK discount factor

Table 3: Scalar parameters. Age-specific parameters are shown in Figure 2.



Note: Labor efficiency e_j , cohort masses $N_j = 1/80$, and survival rates s_j , equal to one before the terminal age $J = 80$. Bottom right: labor income $w e_j$ and consumption c_j in the stationary equilibrium.

Figure 2: Heterogeneous parameters by age.

The steady-state challenge: finding the interest rate that clears. In a representative-agent model, the steady-state interest rate is pinned down by a single Euler equation: $1 = \beta(1 + r)$, hence $r = 1/\beta - 1$, and capital follows from $r = F_K - \delta$. No such shortcut exists here. With 80 cohorts, a constant consumption path would require $\beta(1 + r) = 1$ for every household, but households do not need to hold a constant path: they live a finite life and save for retirement. The Euler equations only give the *growth rate* of each life-cycle consumption profile, $c_{j+1}/c_j = [\beta(1 + r)]^{1/\sigma}$, not the level of r . The interest rate must instead be the price that makes households *willingly* hold the capital stock that firms demand.

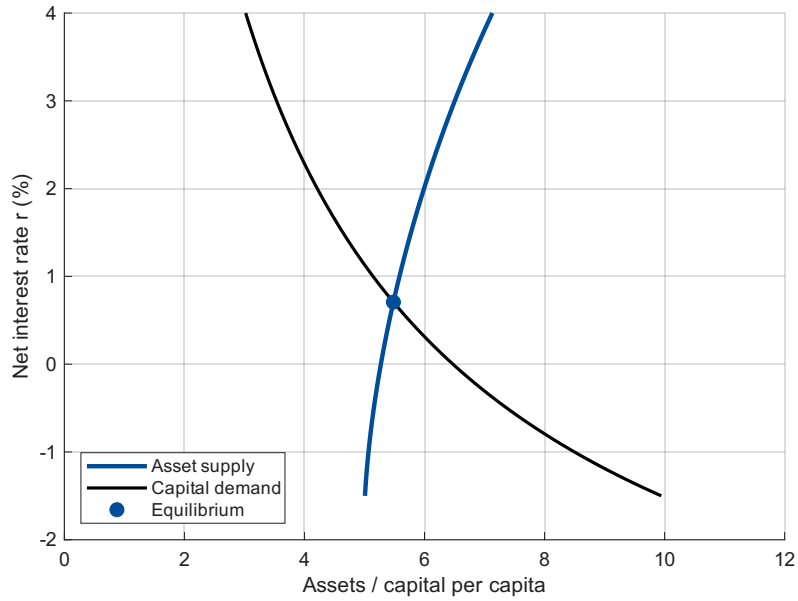
The computation therefore separates the demand and supply schedules of the capital market, both functions of a candidate rate r :

1. *Capital demand schedule.* Firms' first-order condition $r = \alpha(K/\bar{L})^{\alpha-1} - \delta$ gives $K^d(r)$ and the associated wage $w(r)$.
2. *Asset supply schedule.* At $(r, w(r))$, each cohort solves its life-cycle problem: the Euler equations give the consumption growth rate, the lifetime budget gives its level, and the budgets then give the asset path $a_j(r)$. Aggregation gives $A^s(r) = \sum_j N_j a_j(r)$.

The stationary equilibrium is the rate r^* that equates the two,

$$\underbrace{\bar{L} \left(\frac{\alpha}{r^* + \delta} \right)^{\frac{1}{1-\alpha}}}_{\text{capital demand schedule } K^d(r^*)} = \underbrace{\sum_{j=1}^J N_j a_j(r^*)}_{\text{asset supply schedule } A^s(r^*)} . \quad (24)$$

This is one nonlinear equation in one unknown. Figure 3 plots both curves. Demand falls with r : capital is more expensive. Supply rises with r over the relevant range: a higher return makes saving more attractive and, through the lifetime budget, shifts consumption toward old age.



Note: Households' aggregate asset supply $A^s(r)$ and firms' capital demand $K^d(r)$. Their intersection is the stationary equilibrium.

Figure 3: The stationary capital market.

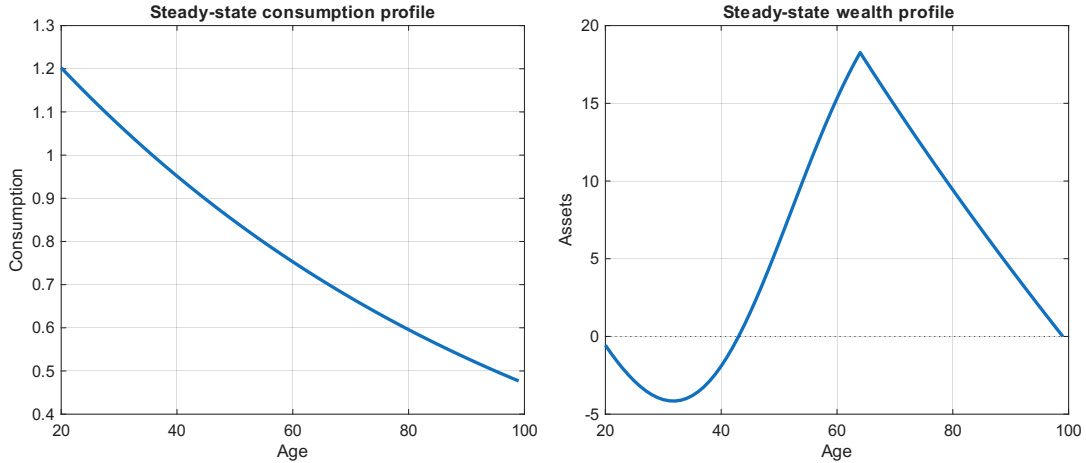
Three features make this step the real difficulty of the steady state. First, $A^s(r)$ has no closed form once households differ: it is a sum of 80 policy functions, each evaluated at the same prices. Second, supply need not be monotone or well defined everywhere: at low rates young households borrow heavily, and a bracketing interval must be chosen where both curves exist. Third, any change in the household problem shifts the whole supply curve, so the steady state must be recomputed.

At the steady state,

$$r^* = 0.00707, \quad K^* = 5.482, \quad w^* = 1.469,$$

which is the intersection of the two curves in Figure 3.

The resulting life cycle. Figure 4 is an *output* of the calibration, not an input: none of these profiles is targeted, and they follow entirely from the parameters of Table 3, the age profiles of Figure 2 and the equilibrium prices. Young households borrow against future labor income, middle-aged households accumulate retirement wealth, and old households decumulate. Negative assets at young ages are therefore not an empirical success but a consequence of the model.



Note: Model output, not a calibration target: these profiles follow from the calibrated parameters and the stationary equilibrium. Negative wealth at young ages is a diagnostic of the model.

Figure 4: Steady-state life-cycle profiles in the one-asset OLG model.

C. Codes: How the Dynare Macro-Processor Works

Why a macro-processor is useful. Writing 80 consumption variables, 79 asset variables, 80 budget equations, and 79 Euler equations by hand would be repetitive and error-prone. Dynare’s macro-processor generates this repeated text before Dynare parses the economic model. It is therefore a *text-generation layer*, not an economic solver and not a MATLAB loop executed during the simulation.

The execution has three stages:

1. the macro-processor reads directives beginning with `@#`, evaluates loops and conditions, and produces an expanded `.mod` file,
2. Dynare parses the expanded variables and equations, checks model consistency, and creates MATLAB functions,
3. the steady-state and perfect-foresight routines solve the resulting nonlinear system.

This distinction is crucial for debugging. A malformed name such as `c_{j}` is a pre-processing issue. A missing equation is a Dynare model issue. A failure to converge is a numerical issue after both earlier stages have succeeded.

Step 1: define the size of the economy. The directive

```
@#define J = 80
```

creates a macro integer named J. It does not create a Dynare parameter and cannot vary endogenously during a simulation. It determines how much source code is generated.

Command-line options can override experiment settings:

```
@#ifndef SHOCK_TYPE
    @#define SHOCK_TYPE = 1
@#endif
```

The `ifndef` guard means: use the default only if the symbol was not supplied on the command line. Consequently,

```
dynare olg80 -DSHOCK_TYPE=2 noclearall
```

runs the second experiment without editing the model file.

Step 2: generate indexed variable names. The declarations use loops:

```
var Z T y w r k C
@#for j in 1:J
    c_{j}
@#endfor
@#for j in 1:J-1
    a_{j}
@#endfor
;
```

The expression `@{j}` substitutes the current macro-loop value into the text. If $J = 3$, the expanded declaration is:

```
var Z T y w r k C c_1 c_2 c_3 a_1 a_2;
```

For $J = 80$, Dynare receives seven aggregate variables, 80 cohort consumptions, and 79 asset positions: $7 + 80 + 79 = 166$ endogenous variables.

Step 3: generate age profiles and conditional statements. The macro-processor can combine loops with conditions:

```
@#for j in 1:J
    N_{j} = 1/@{J};
    @#if j <= 45
        z_{j} = 0.35 + 0.85*exp(-0.5*(({j+19}-50)/14)^2);
    @#else
        z_{j} = 0;
    @#endif
@#endfor
```

The full parameter block sits at the top of each model file, before the `model` block, and is expanded by the macro-processor like any other text. The same block appears in the RANK model, so a change of calibration must be made identically in both files. To change β , σ , δ , or the efficiency profile in an exercise, edit this block in a copy of the model directory and rerun the experiments. The steady-state routines read the resulting `M_.params`. They do not maintain a second numerical calibration.

Cohort index $j = 45$ is age 64. The condition consequently assigns a positive efficiency profile through age 64 and zero labor efficiency from age 65 onward. Again, Dynare never sees an `if` statement at simulation time. It sees 80 explicit parameter assignments.

Step 4: generate aggregators. The aggregation equations (22) of the model are sums over cohorts, so they are written with the same loop syntax:

```
C = 0
@#for j in 1:J
    + N_@{j}*c_@{j}
@#endfor
;
k = 0
@#for j in 1:J-1
    + N_@{j}*a_@{j}
@#endfor
;
```

Starting from zero makes every iteration syntactically identical. The second sum stops at $J - 1$ because the last cohort carries no assets into a next period. For $J = 3$, the macro-processor produces $C = N_1c_1 + N_2c_2 + N_3c_3$ and $k = N_1a_1 + N_2a_2$, the two equations of the model with three cohorts.

Step 5: generate budget and Euler equations. The first and last cohorts are boundary cases and are written separately. Interior budgets are generated by

```
@#for j in 2:J-1
    [name='Budget cohort @{j}']
    c_@{j} + a_@{j}
    = (1+r)*a_@{j-1}(-1) + w*z_@{j};
@#endfor
```

The macro expression `@{j-1}` changes the variable name, while the Dynare timing operator `(-1)` changes the calendar date. These operations must not be confused. For cohort 10, the generated right-hand side contains `a_9(-1)`: the wealth brought by the household that was cohort 9 one period earlier.

Euler equations require both the next age and next calendar date:

```
@#if BC == 1
    c_@{j} = min(c_@{j+1}(+1)/(beta*(1+r(+1)))^(1/sigma),
                (1+r)*a_@{j-1}(-1)+w*z_@{j}+psi_@{j}*T);
@#else
    c_@{j}^(-sigma) = beta*(1+r(+1))*c_@{j+1}(+1)^(-sigma);
@#endif
```

For $J = 80$, the model contains 80 budgets and 79 Euler equations. Together with seven aggregate and shock equations, this gives 166 equations for 166 endogenous variables.

Block	Variables	Equations
Aggregate objects ($Z, \delta_t, Y, w, r, K, C$)	7	7
Cohort consumption ($c_1, \dots, c_{80}$)	80	80 budgets
Cohort assets ($a_1, \dots, a_{79}$)	79	79 Euler equations
Total	166	166

Table 4: Equation counting after macro expansion. Counting before solving is a simple and effective debugging check.

Step 6: select and solve an experiment. Conditional macro directives choose the shock block and output filename:

```
@#if SHOCK_TYPE == 1
    shocks;
    var epsZ;
    periods 1;
    values -0.01;
end;
#else
    shocks;
    var epsD;
    periods 1;
    values 0.02;
end;
#endif

perfect_foresight_setup(periods=100);
perfect_foresight_solver(maxit=200, tolf=1e-9);

@if SHOCK_TYPE == 1
    save('results/results_tfp.mat', 'M_', 'oo_', 'options_');
#else
    save('results/results_dep.mat', 'M_', 'oo_', 'options_');
#endif
```

The first command constructs an initial guess and the deterministic shock path. The second solves all model equations over the full transition horizon. Perfect foresight does not mean that shocks are absent. It means the realized deterministic shock sequence is known when agents choose their transition paths.

The files and the order in which to run them. Each model lives in a single file that reads from top to bottom: calibration, model equations, steady state, shock, simulation, and finally the figures. Only the steady state needs a separate MATLAB function. The baseline uses three files:

- o `olg80.mod` is the 80-cohort model. It also draws the calibration, life-cycle and asset-market figures.
- o `olg80_steadystate.m` computes the stationary equilibrium.
- o `rank80.mod` is the representative-agent benchmark. Its last lines draw the OLG-versus-RANK figure of the same experiment, so it runs after `olg80.mod`.

To reproduce the figures, run in MATLAB `dynare olg80 -DSHOCK_TYPE=1 noclearall`, then `dynare rank80 -DSHOCK_TYPE=1 noclearall`, and repeat both with `-DSHOCK_TYPE=2`. The appendix adds

`olg80_bc.mod`, its steady-state function and one helper, `lifecycle_bc.m`, which solves a single household's life cycle and is shared by both.

How to inspect what the macro-processor produced. When a loop or condition is unclear, copy the model directory and set `J=3` in that copy of `olg80.mod`. Then run:

MATLAB command : expansion only

```
dynare olg80 onlymacro savemacro
```

The option `onlymacro` stops after text expansion, and `savemacro` writes `olg80_macroexp.mod` for inspection. No steady state or transition is solved. This miniature is a reading exercise, not a quantitatively meaningful three-age calibration: retirement, the target cohort, and the life-cycle profile must be redesigned before simulating a shorter life. A three-cohort model should contain $7 + 3 + 2 = 12$ variables and 12 equations. Checking names, boundary conditions, and timing in this small expansion is much easier than debugging the 166-equation version.

D. Simulations

We now use the same economy to ask how cohorts of different ages share the effects of a productivity shock and of a financial crisis.

Perfect-foresight transition paths. The code solves the full nonlinear equilibrium path rather than a first-order local approximation. For each experiment, Dynare simultaneously determines 80 consumption paths, 79 asset paths, capital, output, wages, and interest rates. The terminal part of the 100-period horizon guides the economy back toward its steady state.

A persistent productivity shock. Technology follows

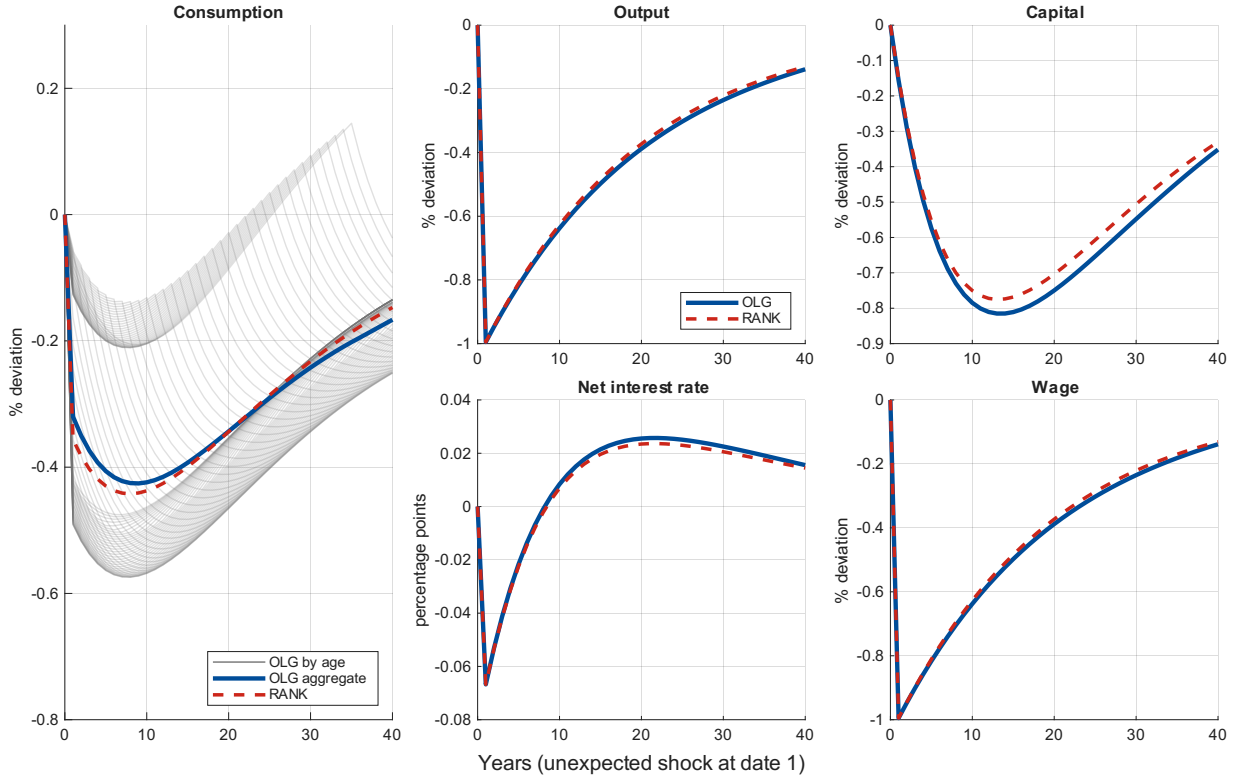
$$\log Z_t = \rho_Z \log Z_{t-1} + \varepsilon_t^Z, \quad \varepsilon_t^Z \sim \mathcal{N}(0, \sigma_Z^2), \quad \rho_Z = 0.9, \quad \sigma_Z = 0.01. \quad (25)$$

At date 1, an unexpected innovation $\varepsilon_1^Z = -\sigma_Z = -0.01$ hits the economy. Output and factor prices move immediately because current production uses the predetermined capital stock K_{t-1} . Consumption falls less than output on impact because households smooth resources over their remaining lives. Capital declines gradually as lower expected returns and income change saving.

A representative-agent benchmark. To see what the cohort structure adds, every simulation is compared with a RANK (here RBC) economy solved by `rank80.mod`. It has the same technology, labor input $\bar{L}$ and TFP process, and a single infinitely lived household whose discount factor $\beta^{RA} = 1/(1 + r^*)$ is chosen so that its steady state reproduces the OLG interest rate. Both economies therefore share K^* , Y^* , w^* and C^* . They differ only in who responds to the shock and how.

The shock is unexpected at date 1, and from then on households know the whole future path of productivity.

Why the OLG and RANK aggregates are so close. Aggregate consumption falls by 0.32% on impact in the OLG and by 0.36% in RANK, with troughs of -0.43% and -0.44% . Capital falls by at most 0.82% and 0.78%. This similarity is not a coincidence. Both economies start from the same steady state. Labor is inelastic and capital predetermined, so output is identical on impact and only the split



Note: OLG (blue) and RANK with the same steady state (dashed). Left: each grey line is the consumption of one age group $c_{j,t}$. The thick line is their population-weighted mean, aggregate consumption C_t .

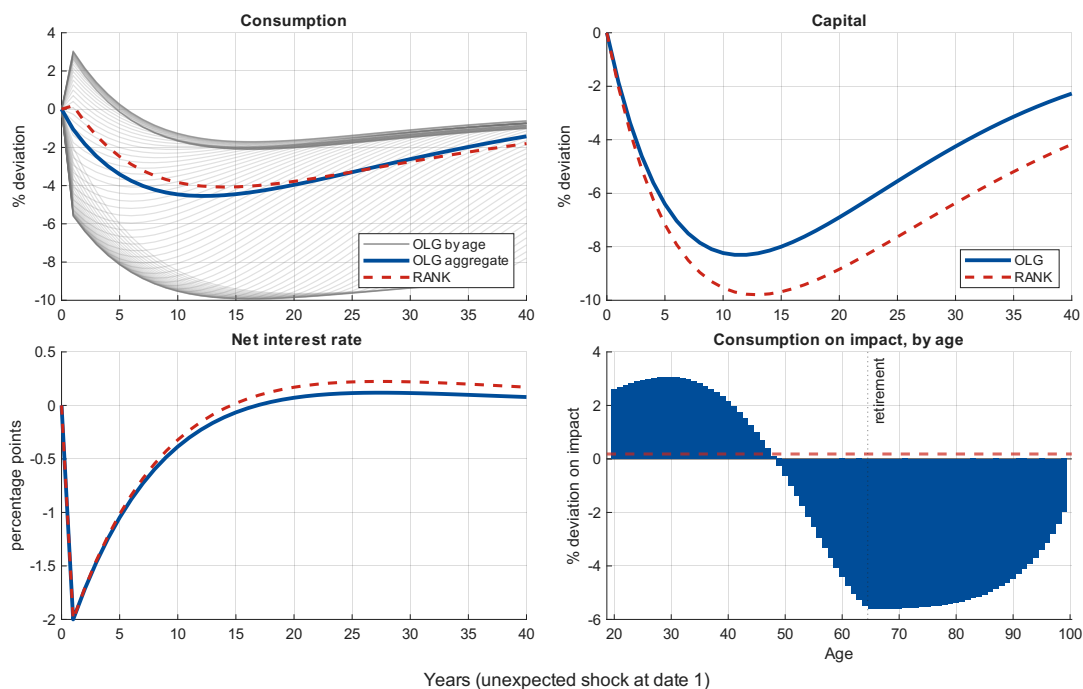
Figure 5: IRF to a negative productivity shock.

between consumption and saving can differ. Every cohort smooths consumption like the representative household. Heterogeneity changes aggregates only if the households most exposed to the shock also have different propensities to consume. Here, workers bear the wage loss but, like the representative household, have a low MPC.

The cohort lines show what the aggregate hides. On impact, households aged 20–49 cut consumption by 0.42% to 0.49% because the lower wage reduces their remaining labor income. Retirees live on accumulated assets and move by only 0.06% to 0.13%. Their consumption falls later, as the cohorts that lost labor income age into retirement with less wealth.

A financial crisis: a depreciation shock. The second experiment hits the return on capital rather than productivity. At date 1, the depreciation rate rises unexpectedly by two percentage points, $\varepsilon_1^D = 0.02$, and then decays with persistence $\rho_D = 0.9$. Output is unchanged on impact, but the net return $r_t = \alpha Y_t / K_{t-1} - \delta_t$ falls by about two percentage points: part of the wealth households hold is lost. In RANK, the representative household smooths this loss over an infinite horizon and its consumption barely moves on impact. In the OLG, the same loss falls on households with very different horizons.

The two models now diverge. Aggregate consumption falls by 1.05% on impact in the OLG and rises by 0.18% in RANK. The age profile explains why. Retirees hold the most assets relative to their remaining resources and have short horizons, so the lower return and the capital loss cut their consumption sharply, by up to about 5.5% around age 70. Young workers own little wealth and gain from a lower interest rate, since their human wealth is discounted less: they raise consumption by up to about 3%. The retirees' MPC out of wealth is high and their weight in aggregate consumption is large, so the net effect is negative. Capital falls by up to 8.3% in the OLG against 9.8% in RANK, because the OLG cohorts that gain, the young, are the ones who save. The shock is a pure change in the return on wealth, and it creates an



Note: Top left: consumption by age (grey), aggregate consumption (thick) and RANK (dashed). Top right and bottom left: capital and interest rate. Bottom right: consumption on impact by age.

Figure 6: IRF to a positive depreciation shock (financial crisis).

aggregate response only because households differ by age.

Interpretation and limitations. The simulations establish three results. First, a common aggregate shock generates different cohort responses because resources and horizons vary by age. Second, the interest rate is part of the equilibrium response rather than an exogenous discounting device. Third, a shock to the return on wealth changes aggregate dynamics when it falls on households with different horizons and MPCs.

These experiments still do not reproduce the full distribution of liquid and illiquid wealth. The model has only one asset, no mortality risk before the terminal age, and no idiosyncratic employment transitions within age. There are also no nominal rigidities or monetary policy. The model is therefore an OLG laboratory for life-cycle saving and capital-market clearing, not a HANK model and not a quantitative replication of [Kaplan et al. \(2014\)](#).

PART III : Code Exercises

The exercises use the supplied files. The objective is to produce and read graphs, not to write new code. Work on a copy of the `model` directory.

Exercise 1: Reading the perfect-foresight solution

After `perfect_foresight_solver`, Dynare stores the whole solution in the structure `oo_`. Two matrices matter. The matrix `oo_.endo_simul` has one row per endogenous variable, in the order

of `M_.endo_names`, and one column per date. Column 1 is the initial condition, columns 2 to $T+1$ are the simulated periods and column $T+2$ is the terminal condition. The matrix `oo_.exo_simul` is the transpose in spirit: one row per date and one column per exogenous variable, in the order of `M_.exo_names`. The vector `oo_.steady_state` gives the stationary value of each endogenous variable, in the same order as the rows of `oo_.endo_simul`.

1. Run `dynare olg80 -DSHOCK_TYPE=1 noclearall` and display `size(oo_.endo_simul)` and `size(oo_.exo_simul)`. Why are there $T+2 = 102$ columns and rows although the model is simulated over $T = 100$ periods? In which row and column of `oo_.exo_simul` is the productivity innovation?
2. Find the row of output in `oo_.endo_simul` with `strcmp(cellstr(M_.endo_names), 'y')`. Plot the percentage deviation of output, capital and the interest rate from their steady state, $100(x_t/x^* - 1)$, over the first 40 dates. Why does the interest rate deserve a deviation in percentage points instead?
3. Read `oo_.steady_state` to report r^* and K^* . Compare it with the first and the last columns of `oo_.endo_simul`. What do these two comparisons tell you about the initial and terminal conditions of the simulation?
4. Simulate by hand. Store the solution, then run `oo_ = perfect_foresight_setup(M_, options_, oo_)`, set `oo_.exo_simul(2,1)=0.01` and run `oo_ = perfect_foresight_solver(M_, options_, oo_)`. Plot the output response against the one of question 2. Is it the mirror image? Then put the shock -0.01 at date 11 (row 12) instead and solve again. What happens to output at dates 1 to 10, and why?

Exercise 2: Plots and an alternative calibration

1. Run `olg80.mod` and then `rank80.mod` with `-DSHOCK_TYPE=1`. Describe the steady-state consumption and asset profiles by age and the response of aggregate consumption to the TFP shock in the OLG and in RANK.
2. In the parameter block of `olg80.mod` and `rank80.mod`, lower the discount factor from $\beta = 0.97$ to $\beta = 0.96$ and rerun. Report r^* and K^* in both calibrations. Why does the interest rate rise?
3. Plot the two asset profiles on the same graph. Which ages save less?
4. Compare the consumption responses to the TFP shock in both calibrations, for a young worker, a retiree, and the aggregate. Is the gap with RANK larger or smaller?

Exercise 3: A more persistent financial crisis

1. Run `dynare olg80 -DSHOCK_TYPE=2 noclearall` and reproduce Figure 6.
2. Using Figure 6, explain why a depreciation shock lowers aggregate consumption on impact in the OLG but not in RANK.
3. In `olg80.mod`, raise the persistence from $\rho_D = 0.9$ to $\rho_D = 0.99$ and rerun. Since $0.99^{100} \approx 0.37$, also extend the horizon to `periods=400`. Plot capital and the interest rate for both values on

the same graph. Why is the fall in capital larger and longer?

4. Plot consumption on impact by age for both values. Which cohorts gain, which lose, and why?
5. In RANK, consumption barely moves on impact but falls later. Why? Which feature of the OLG would you remove to make its aggregate response close to RANK again?

Appendix

A. The Borrowing Limit

Everything above assumes that a young household can borrow against its future wages. This appendix removes that assumption and nothing else. It first states the restriction and its optimality conditions, then shows how the model files implement it, and finally studies how it changes the marginal propensity to consume and the response to the productivity shock.

The ASV borrowing limit. The baseline allows a young household to consume against earnings that it will receive much later. But a bank need not accept future wages as collateral. What happens if the household must finance current consumption only from current income and accumulated assets? To answer this question, we retain the baseline as a benchmark and add the restriction

$$a_{j,t} \geq 0, \quad j = 1, \dots, J-1. \quad (26)$$

This is the no-borrowing restriction implemented in the ASV research file `ASV_bc_ss.mod`, in its Euler block. We extract the restriction alone: the research file also includes portfolio adjustment costs, fiscal wedges, and mortality, whereas the additional teaching case keeps $s_j = 1$ before the terminal age and sets those extra mechanisms aside. We therefore reproduce a mechanism from ASV, not its full quantitative calibration.

Let $m_{j,t}$ denote cash on hand, the right-hand side of the household budget. Thus $m_{1,t} = w_t e_1 + \psi_1 \mathcal{T}_t$ and $m_{j,t} = (1 + r_t)a_{j-1,t-1} + w_t e_j + \psi_j \mathcal{T}_t$ at subsequent ages. Since $a_{j,t} = m_{j,t} - c_{j,t}$, the restriction is equivalent to $c_{j,t} \leq m_{j,t}$. The optimality conditions become

$$c_{j,t}^{-\sigma} = \beta(1 + r_{t+1})c_{j+1,t+1}^{-\sigma} + \lambda_{j,t}, \quad (27)$$

$$\lambda_{j,t} \geq 0, \quad a_{j,t} \geq 0, \quad \lambda_{j,t}a_{j,t} = 0. \quad (28)$$

The multiplier $\lambda_{j,t}$ measures the marginal value of relaxing the borrowing limit. When assets are strictly positive, $\lambda_{j,t} = 0$ and the original Euler equation holds. When the household would like to borrow, marginal utility today exceeds the discounted marginal utility of future consumption: saving less would be desirable, but is infeasible.

Rearranging the unconstrained Euler equation gives $c_{j,t}^E = c_{j+1,t+1}/[\beta(1 + r_{t+1})]^{1/\sigma}$. Combining the two regimes yields the compact ASV-style condition

$$c_{j,t} = \min \left\{ \frac{c_{j+1,t+1}}{[\beta(1 + r_{t+1})]^{1/\sigma}}, m_{j,t} \right\}. \quad (29)$$

This is an equilibrium equation: future consumption is solved jointly with current consumption and assets. It is not a prescription to clip negative assets after solving the unconstrained model. Clipping would violate household budgets and asset-market clearing. At age J , the terminal budget still requires consumption of all remaining resources.

Adding the ASV limit without rewriting the whole model. The appendix file `olg80_bc.mod` preserves the baseline budgets, production, and aggregation. Its macro switch `BC` selects which optimality condition is generated. With `BC=0`, the original Euler equation is retained. With `BC=1`, the Euler loop implements (29). For an interior cohort, the essential lines are

```
@#if BC == 1
    c_@{j} = min(c_@{j+1}*(1)/(beta*(1+r*(+1)))^(1/sigma),
                (1+r)*a_@{j-1}*(-1)+w*z_@{j}+psi_@{j}*T);
@#else
    c_@{j}^(-sigma) = beta*(1+r*(+1))*c_@{j+1}*(+1)^(-sigma);
@#endif
```

In the actual file, the inherited-wealth term $(1+r)a_{j-1}$ is dropped for the newborn by an extra condition on $j > 1$, because there is no inherited a_0 . No multiplier variable is needed in this representation, so both variants still have 166 equations and 166 endogenous variables. The multiplier is recovered afterwards from the Euler wedge in (27), allowing us to verify all three complementary-slackness conditions.

The steady-state file must change as well. The baseline's single lifetime-budget formula assumes that borrowing is feasible at every age. The routine `lifecycle_bc.m` instead computes optimal consumption from the tightest remaining prefix budget: consumption today must be consistent with nonnegative assets at every intervening age, not only with zero assets at death. At constant prices, let h count years from the current age, y_h denote the remaining income sequence, a^- inherited assets, $R = 1 + r$, and $g = (\beta R)^{1/\sigma}$. Current optimal consumption is

$$c_0 = \min_{n=0,\dots,H-1} \frac{Ra^- + \sum_{h=0}^n y_h / R^h}{\sum_{h=0}^n (g/R)^h}, \quad (30)$$

where H is remaining lifetime. Without the borrowing limit, only $n = H - 1$ is retained. With the limit, the minimizing prefix identifies the first date at which assets run out. After the current choice, the routine updates assets and repeats for the next age. The steady-state function `olg80_bc_steadystate.m` calls this routine inside an outer loop that adjusts r until aggregate household assets equal firms' capital demand. This explains why simply reusing the baseline steady state would fail.

To reproduce the MPC profiles and the general-equilibrium comparison, run

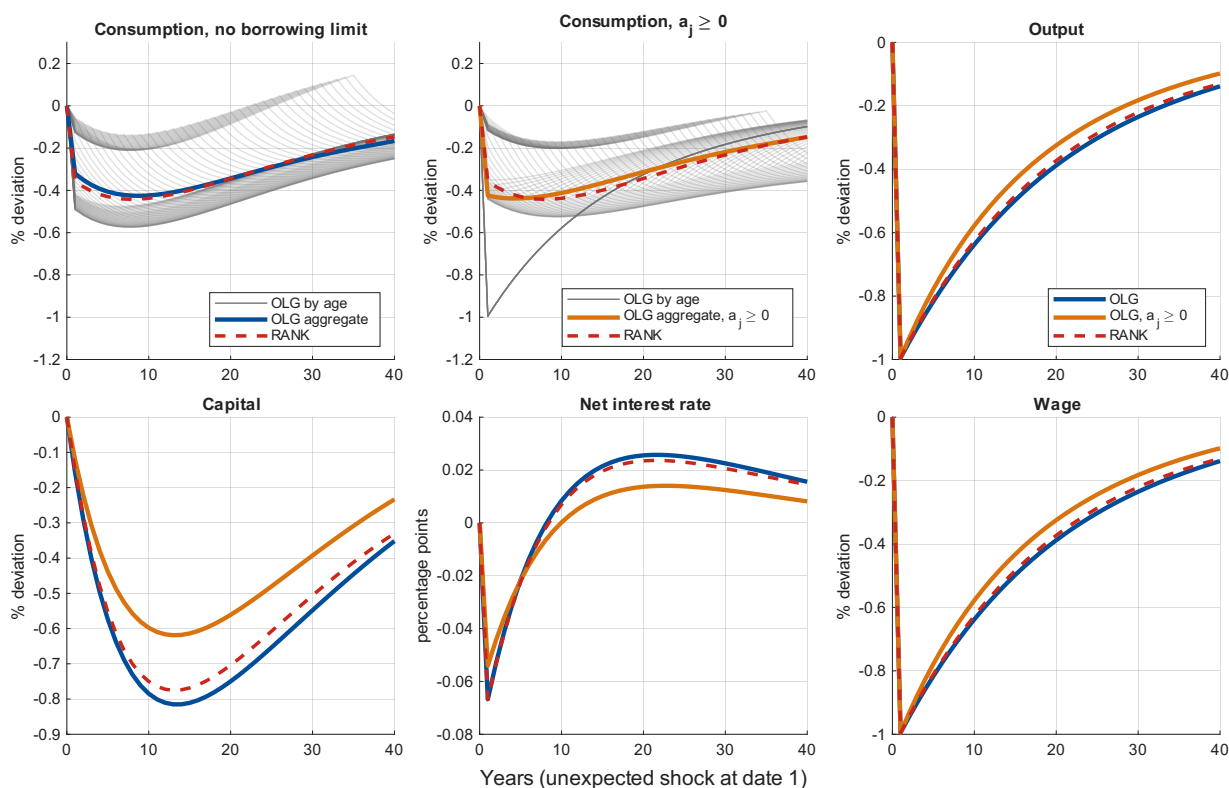
MATLAB command : from the model directory

```
dynare olg80_bc -DBC=1 -DSHOCK_TYPE=1 noclearall
dynare olg80_bc -DBC=0 -DSHOCK_TYPE=2 -DTARGET=6 noclearall
dynare olg80_bc -DBC=1 -DSHOCK_TYPE=2 -DTARGET=6 noclearall
```

The first command computes the MPC profiles and draws Figures 8 and 7; it needs the baseline TFP runs of `olg80.mod` and `rank80.mod` first. The next two print the general-equilibrium response to a windfall in each regime, recomputing the stationary equilibrium each time. Replace `TARGET=6` by `TARGET=56` for age 75. The nonlinear `min` specification follows the ASV implementation. A solver success message alone

is insufficient. For harder cases, Dynare also offers mixed-complementarity solvers, but these require a different equation specification rather than simply adding a solver option to this file.²

The productivity shock with a borrowing limit. Figure 7 repeats the productivity experiment of Section D with the limit $a_j \geq 0$ (orange). Households aged 20–33 are constrained and consume their current income: their consumption falls one for one with the wage, by 1%, instead of 0.4%–0.5% elsewhere. Aggregate consumption falls more on impact (-0.43%), but the trough is almost unchanged and capital falls less (-0.62%). The constrained households are young and earn little, so their weight in aggregate consumption is small, and they hold no assets that could be run down. A borrowing limit on a small, low-income group creates large individual differences but only a modest aggregate effect. Larger aggregate effects require a larger mass of constrained households exposed to the shock: this is the logic of the HANK models of Session 3.



Note: OLG without borrowing limit (blue), OLG with $a_j \geq 0$ (orange) and RANK with the same steady state as the unconstrained OLG (dashed). Top left and centre: each grey line is the consumption of one age group $c_{j,t}$. The thick line is their population-weighted mean, aggregate consumption C_t .

Figure 7: The productivity transition with and without the borrowing limit.

Additional case: borrowing limits and MPCs. Imagine two 25-year-old households with the same preferences and earnings prospects. One can borrow against future wages, while the other cannot. The first can already smooth consumption and will save most of a small unexpected transfer. The second may be consuming its entire current income and would like to borrow more. If the transfer is small enough that the limit still binds, it consumes the whole amount. Age alone did not create this difference: access to credit changed the household's feasible choices.

²See the [Dynare Reference Manual](#), deterministic simulation and `lmmcp`. The original OLG validation used Dynare 7.0. This standard perfect-foresight model does not use the HA complementarity block. For the whole course install Dynare 7.x, at least 7.1 (7.2 recommended).

We define the household impact MPC by a one-sided windfall experiment,

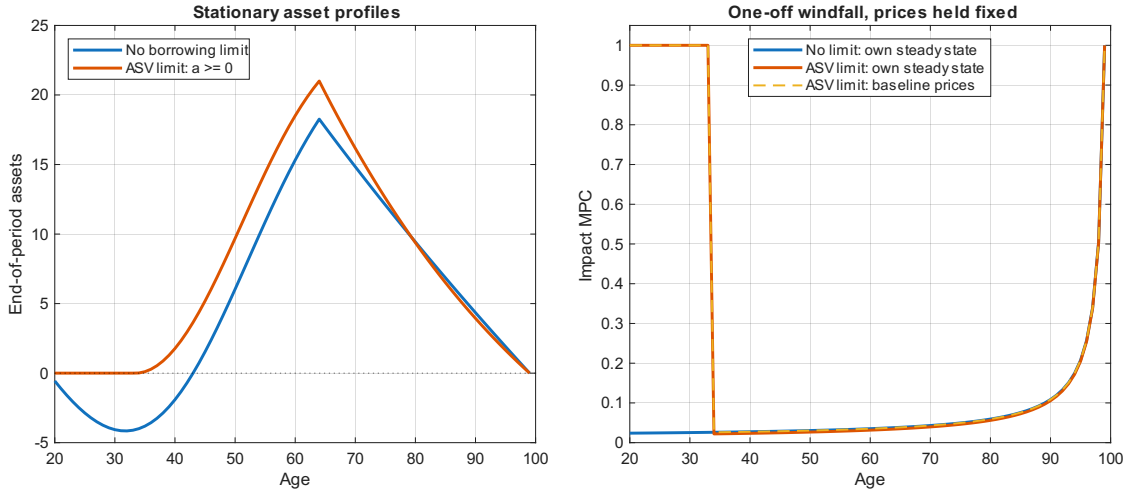
$$\text{MPC}_j^{PE} = \lim_{\epsilon \downarrow 0} \frac{c_j(a_j^-, y_j + \epsilon; \{y_{j+h}\}_{h \geq 1}, r, w) - c_j(a_j^-, y_j; \{y_{j+h}\}_{h \geq 1}, r, w)}{\epsilon}. \quad (31)$$

Inherited assets, future earnings, and prices are held fixed within each comparison. The windfall is received unexpectedly at the current age and does not recur. This is a partial-equilibrium derivative for an individual of negligible mass. No aggregate fiscal experiment is asserted. In the code, $\epsilon = 10^{-4}$, and dividing it by ten changes the estimated MPC by less than 4×10^{-11} .

For an unconstrained household with $H = J - j + 1$ remaining periods at constant prices, consumption growth is $g = (\beta R)^{1/\sigma}$. A one-unit addition to present-value wealth is allocated across the remaining consumption path. Substitution into the lifetime budget gives

$$\text{MPC}_j^{PE} = \left[\sum_{h=0}^{H-1} (g/R)^h \right]^{-1}. \quad (32)$$

If instead the household remains strictly constrained, $a_j = 0$ and $c_j = m_j$, hence $\text{MPC}_j^{PE} = 1$. At the terminal age the MPC is also one, but for a different reason: there is no remaining consumption date. At regime boundaries a positive and a negative windfall need not have the same derivative. Our reported MPC uses the positive direction.



Note: Left: stationary asset profiles. Right: household MPCs to an unexpected windfall, holding prices fixed within each experiment. Solid curves use each model's own stationary equilibrium. The dashed curve recomputes the constrained life cycle at the baseline prices to separate the credit restriction from the change in equilibrium prices.

Figure 8: The ASV-style borrowing limit in the teaching OLG.

The left panel of Figure 8 identifies the mechanism behind the right panel. Assets are zero from ages 20 to 33 in the constrained economy, and these households consume the marginal windfall in full. Once households save strictly positive amounts, they return to consumption smoothing. Near the end of life the remaining horizon shrinks, so the MPC rises again. This produces a U-shaped profile with a sharp transition out of the constrained regime. The sharpness reflects the deliberately deterministic, one-type-per-age environment, not a claim that empirical age profiles have such a discontinuity.

Why does the borrowing limit not raise every MPC? Removing young households' borrowing changes net asset supply and the stationary capital stock. Here K^* rises and the net return falls. At saving ages,

Object	No borrowing limit	ASV limit $a_j \geq 0$
Stationary capital K^*	5.4817	7.5038
Annual net interest rate r^*	0.7072%	-0.5653%
Household MPC at age 25	0.02461	1.00000
Household MPC at age 35	0.02637	0.02227
Household MPC at age 75	0.04964	0.04623
GE transfer-response ratio at age 25	0.02495	1.00000
GE transfer-response ratio at age 75	0.04964	0.04621

Table 5: Recomputed equilibria and matched windfall experiments. Household MPCs hold prices fixed at each economy's own steady state. GE ratios use an unexpected balanced transfer at the first simulation date and allow all equilibrium prices to adjust.

equation (32) then gives a somewhat lower MPC. The dashed fixed-price curve makes this distinction visible: in this deterministic calibration, once the household no longer faces a current or future binding limit, the MPC coincides with the unconstrained formula at the same prices. The lower solid constrained curve at saving ages is therefore a price effect, not evidence that a binding borrowing restriction directly reduces the MPC. The negative equilibrium net return is a consequence of the unchanged illustrative calibration, and neither the rate nor the ages of constraint are empirical targets.

For the GE comparison, `olg80_bc.mod` uses a balanced windfall, a transfer between cohorts, instead of the depreciation shock. The parameter `TARGET` is the index of the cohort that receives it (age = `TARGET`+19, so 6 is age 25 and 56 is age 75). The recipient gets a one-off windfall financed by an equal tax on all other cohorts, so the transfer sums to zero and creates no aggregate resources. The innovation is set to 10^{-4} at simulation date 1. Inherited assets are fixed at the pre-shock steady state, so households have no simulated period in which to anticipate and save the transfer. Thereafter they correctly foresee the deterministic equilibrium path. The recipient receives $(1 - 1/J)10^{-4}$ net of its own tax, and this is the denominator of the reported ratio. Responses are measured against each model's own steady state.

Result. A borrowing limit can create high MPCs among young households, while the equilibrium-price adjustment changes MPCs at saving ages. In the productivity transition it makes young cohorts adjust one for one with the wage, yet the aggregate effect stays modest because these cohorts are small in consumption terms.

Exercise 4: Borrowing limits and the MPC

1. Run `dynare olg80_bc -DBC=1 -DSHOCK_TYPE=1 noclearall` and reproduce Figure 8. At which ages is the borrowing limit binding?
2. Why is the MPC equal to one at age 25 in the constrained economy? Why is it also equal to one at age 99? Is the cause the same?
3. Why is the MPC at saving ages slightly lower with the borrowing limit? Use the dashed fixed-price curve to answer.

Common Notation Across the Three Sessions

The same economic object has the same symbol in all three handouts. Indices may be suppressed for a generic household. In the OLG model j denotes age; in Markov matrices j, k label earnings states. A stationary object has no time subscript or a bar.

Symbol	Common meaning and timing
$c_{i,t}, C_t$	Individual and aggregate consumption at date t .
$a_{i,t}$	Assets chosen at the end of t ; inherited wealth is $a_{i,t-1}$. Dynare: a , a(-1) . An age- j household inherits $a_{j-1,t-1}$.
$a, a', g_t(a, e)$	In a recursive problem, a is inherited wealth; $a' = g_t(a, e)$ is the end-of-period choice. Thus $a = a_{t-1}$ and $a' = a_t$, not an alternative calendar-time convention.
A_t, K_t, B_t	End-of-period aggregate household assets, productive capital and public debt. Current output uses K_{t-1} . Clearing: $A_t = K_t + B_t$ when both assets exist; $A_t = K_t$ in Sessions 1–2 and $A_t = B_t$ in Session 3.
D_t, N_j	Beginning-of-period state distribution, of total mass one; N_j are OLG cohort masses. In HA models D_t is over (a_{t-1}, e_t) .
e, Z_t, z_t	Individual labor efficiency; aggregate TFP level; aggregate log-TFP deviation when used, $Z_t = \bar{Z} \exp(z_t)$.
L_t, N_t	Effective labor and common hours. Sessions 1–2 fix effective labor; Session 3 has $L_t = N_t$ because mean efficiency is one.
r_t, R_t, w_t	Net real return, gross real return $R_t = 1 + r_t$, and real wage per efficiency unit.
i_t, π_t	Nominal rate contracted at t and net inflation. In Session 3, $1 + r_t = (1 + i_{t-1})/(1 + \pi_t)$.
β, σ, λ	Discount factor, CRRA/inverse EIS, and nonnegative borrowing-constraint multiplier. Values need not match across calibrations.
$\Pi, \bar{p}$	Row-stochastic earnings transition matrix and invariant column probabilities, $\bar{p} = \Pi^\top \bar{p}$. Inflation is π_t , not $\bar{p}$.
$G_t, T_t, \mathcal{T}_t$	Government purchases, taxes, and transfer/redistribution scale. Session 1 uses signed balanced age loadings; Session 3 records gross transfers and taxes separately.

Code aliases are not new economic objects. The validated OLG code retains **A** for Z , **z_j** for e_j , and **T** for $\mathcal{T}$. Session 2 uses **Zbar*exp(z)** for Z_t ; Session 3 uses **Tax** for T_t and **Tr** for $\mathcal{T}_t$. Normann’s appendix notation maps **eis** to $1/\sigma$ and **frisch** to $1/\nu$. Keeping these aliases explicit preserves the tested code without changing their meaning in the equations.

Units and experiments. Session 1 is annual, Session 2 uses illustrative model periods, and Session 3 is quarterly. A fixed-price cash-windfall MPC is not a general-equilibrium fiscal multiplier. A percentage change in a quantity is not a percentage-point change in a rate. The OLG TFP process is in levels; the HA-RBC process is in log deviations. Different calibrations and shock processes are intentional, not notation inconsistencies.

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